

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Dept. of Metallurgical and Materials Engineering

2. **Subject Name:** Physics of Materials

L-T-P: 3-0-0

Credits: 3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

(1) Classical treatment of electrons in solids: Electron as a particle, Kinetic theory of gasses, Drude's model for electrical & thermal conductivity, Classical interpretation of Hall effect, Hall factor, Successes and limitations of Drude's model.

(2) Dielectric solids: Concept of relative permittivity of solids, Static dielectric constant and its measurement, Polarization and its types, Internal fields in solids and liquids, Ferroelectricity (Barium Titanate) and piezoelectricity.

(3) Introduction to quantum mechanics: History of quantum mechanics, Double-Slit Experiment, De Broglie's Hypothesis, Superposition, Heisenberg's Uncertainty Principle, Properties of wave functions, Time-independent Schrödinger equation, Quantum operators, Fermi-Dirac statistics, Drude Sommerfeld model, Fermi energy, Fermi surface, Density of states, Specific heat of electrons.

(4) Electronic properties of crystals: Anisotropic potential, Wigner-Seitz cell, Reciprocal space, Brillouin Zones, and Bragg plane, Quantization in a 1-D lattice, Origin of Bands, Metal, Semiconductor and Insulator, Kronig-Penney Model, Free electron approximation and tight binding.

(5) Physics of semiconductors: Direct vs indirect bandgap, Carrier concentration in intrinsic and doped semiconductors, Junction potential.

(6) Lattice vibrations: 1D monoatomic chain and dispersion relation, 1D diatomic chain and dispersion relation, Phonons, Classical theory of lattice heat capacity, Einstein's theory of lattice heat capacity, The Debye approximation and shortcomings.

Reference Books:

- (1) Kittel, Charles, and Paul McEuen. *Introduction to solid state physics*. John Wiley & Sons, 2018.
- (2) Dekker, Adrianus J. *Electrical engineering materials*. Prentice-Hall, (1959).
- (3) Solymar, Laszlo, Donald Walsh, and Richard RA Syms. *Electrical properties of materials*. Oxford university press, 2014.

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Classical treatment of electrons in solids	4
2.	Dielectric solids	5
3.	Gentle introduction to quantum mechanics	10
4.	Electronic properties of crystals	7
5.	Gentle introduction to the physics of semiconductors	5
6.	Lattice vibrations	5
Total number of hours		36